Howard Sun | Software Engineer

646.801.8812 | Connecticut/NYC · Open to relocation | howardsun@pm.me
howardsun.me | github/howardsun-dev | linkedin/howardsun-swe

SUMMARY

Full-stack software engineer focused on building reliable, well-tested applications with TypeScript, React, Node.js, and modern web tooling. Recent experience contributing to Swell, an open-source API testing platform, with work spanning security hardening, test infrastructure, Redux modernization, and frontend architecture. Previously co-founded and led engineering for a children's coding education platform, building production systems across AWS, Node.js, TypeScript, and database infrastructure.

SKILLS

Languages: TypeScript, JavaScript (ES6), Python, C++
Frontend: React, Redux Toolkit, Zustand, TanStack Router, React Router, HTML5, CSS3 (Sass), Tailwind CSS, Material UI, DaisyUI, Chart.js, webpack, Vite
Backend: Node.js, Express, WebSocket, Socket.IO, REST APIs, JWT/Auth, Cloudinary, Arcjet
Desktop & Tooling: Electron, electron-builder, CLI tools, Git, Figma
Databases: PostgreSQL, MongoDB , Amazon Aurora
Infrastructure & Cloud: AWS EC2, AWS S3, AWS Elastic Beanstalk, AWS VPC, CloudFront, ELB/ALB, Docker, GitHub Actions
Testing: Jest, Playwright, Mocha, Chai, Vitest

EXPERIENCE

Swell · Software Engineer — April 2023 – February 2025

- Implemented Content Security Policy (CSP) in Swell using nonce and SHA-256-based hashing to secure backend protocol testing, mitigate cross-site scripting (XSS) and code injection attacks, and protect user data without impacting application performance
- Reduced technical debt by refactoring legacy Redux architecture: migrated to Redux Toolkit, implemented type-safe hooks, expanded TypeScript coverage, removed deprecated functions and standardized state management to improve maintainability and code quality
- Developed a React-based binary file upload feature using Test-Driven Development (TDD) with Jest, enabling validation of backend server routes across different file types and payload sizes to expand API testing capabilities
- Increased overall test coverage by 15% by repairing failed tests and authoring new unit and integration tests using Jest, Mocha, Chai and Playwright; improved test reliability with Jest mocking and Chai semantic assertions
- Improved application reliability and stability by identifying and resolving security vulnerabilities in API Simulation Tool through root cause analysis and Node.js debugging, preventing crashes and runtime failures.
- Standardized CSS and Material UI theming across the application by unifying CSS classes, implementing Material UI theming architecture, and adding dark mode support to ensure a consistent and accessible user experience.

Mission Coders, LLC · CTO, Co-Founder — May 2018 – December 2022

- Transitioned from an on-premise monolithic server to a scalable AWS infrastructure with Elastic Load Balancing (ELB-ALB) and Elastic Compute Cloud (EC2), reducing resource overhead by 40%, enabling auto-scaling based on traffic patterns, and improving reliability and cost-efficiency across multiple regions
- Designed and implemented a scalable database solution using Amazon Aurora RDS to manage student and course registration; utilized Aurora's auto-scaling capabilities and read replicas to accommodate fluctuating traffic and high read workloads and achieving 99% availability
- Led a six-member team to develop a children's programming learning platform using JavaScript, Node.js, Express, TypeScript, and Redux; integrated Scratch to deliver an engaging, interactive educational tool, completing the project 5 weeks ahead of the deadline

Quark Virtual Services · Technical Manager, Co-Founder — April 2013 – October 2022

- Streamlined deployment for managed and unmanaged Virtual Private Servers (VPS) using KVM and OpenVZ; applied infrastructure-as-code principles and configuration management (Ansible, Puppet) to create reusable, version-controlled templates, reducing manual effort and overall deployment time by 15%

PROJECTS

LANShare (Web & Electron) | https://github.com/howardsun-dev/lanshare-electron

- Built a TypeScript/Node.js LAN file-sharing application with Electron and Express, featuring secure traversal-safe file serving, browser-based controls, CLI support, automated CI testing pipelines, and lightweight local-network deployment for trusted environments.

QuickChat – Real-time chat platform | github.com/howardsun-dev/quickchat

- Built a deployed real-time messaging platform using React, Node.js, Express, MongoDB, and Socket.IO with authentication, persistent chat data, and online/offline presence detection. Implemented Zustand for state management, optimistic UI updates, secure media uploads through Cloudinary, and request protection/rate limiting with Arcjet. Designed a responsive Tailwind CSS/DaisyUI interface for desktop and mobile users.

Home Automation – Engineering Project

- Built a Raspberry Pi and Arduino-based home automation system using C++ for low-level device control. Integrated subsystems for automated blind control, timing-based routines, and PWM-driven hardware behavior. Optimized memory usage and control logic for reliable long-running operation.

EDUCATION

Central Connecticut State University – B.S. Electronic Technology — 2012

INTERESTS

Diving into Korean and Russian languages; geeking out over all things Sci-Fi, especially Dune or Star Trek; exploring new cities and embracing nature while camping